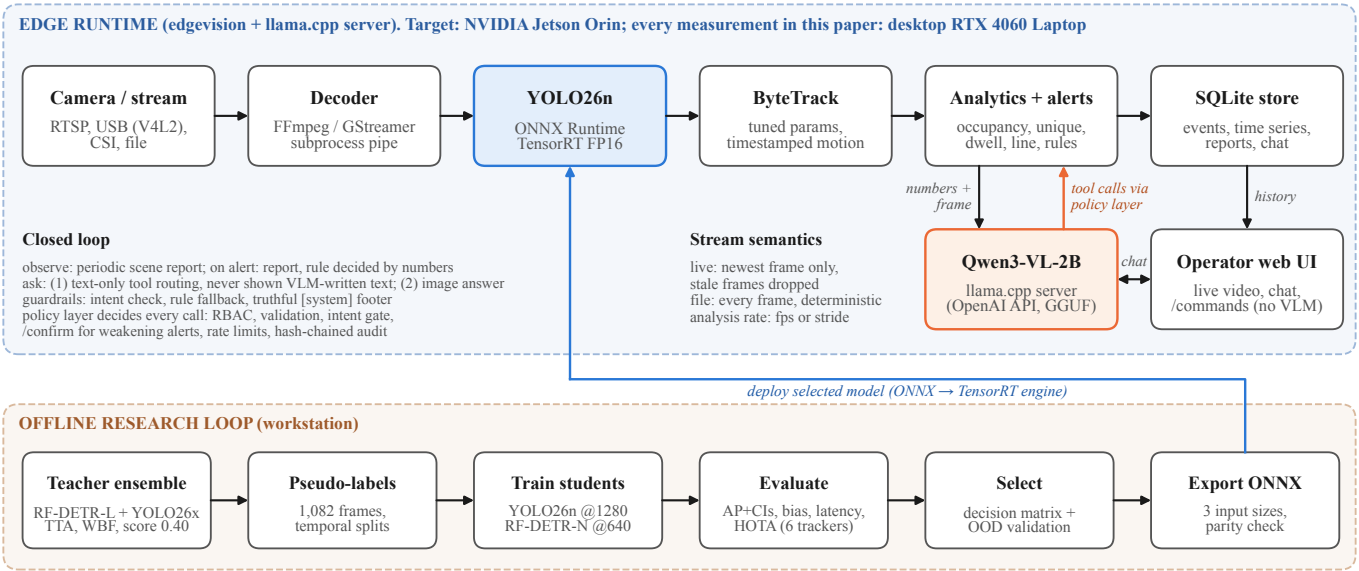

EDGE RUNTIME (edgevision + llama.cpp server). Target: NVIDIA Jetson Orin; every measurement in this paper: desktop RTX 4060 Laptop
Camera / stream
RTSP, USB (V4L2), CSI, file
Decoder
FFmpeg / GStreamer subprocess pipe
YOLO26n
ONNX Runtime TensorRT FP16
ByteTrack
tuned params, timestamped motion
Analytics + alerts
occupancy, unique, dwell, line, rules
SQLite store
events, time series, reports, chat
numbers + frame
tool calls via policy layer
history
Closed loop
observe: periodic scene report; on alert: report, rule decided by numbers
ask: (1) text-only tool routing, never shown VLM-written text; (2) image answer
guardrails: intent check, rule fallback, truthful [system] footer
policy layer decides every call: RBAC, validation, intent gate, /confirm for weakening alerts, rate limits, hash-chained audit
Stream semantics
live: newest frame only, stale frames dropped
file: every frame, deterministic
analysis rate: fps or stride
Qwen3-VL-2B
llama.cpp server (OpenAI API, GGUF)
chat
Operator web UI
live video, chat, /commands (no VLM)
deploy selected model (ONNX → TensorRT engine)
OFFLINE RESEARCH LOOP (workstation)
Teacher ensemble
RF-DETR-L + YOLO26x TTA, WBF, score 0.40
Pseudo-labels
1,082 frames, temporal splits
Train students
YOLO26n @1280 RF-DETR-N @640
Evaluate
AP+CIs, bias, latency, HOTA (6 trackers)
Select
decision matrix + OOD validation
Export ONNX
3 input sizes, parity check